Advanced Mathematical Techniques Worksheet 6
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The purpose of this worksheet is to give you some practice working with the third integral representation of Beta function as well as introducing and using regulators to determine the value of an integral. You may use Mathematica or another computer algebra system to check your work, but please do your work by hand.

Use whichever means necessary to determine the value of the following integrals. If the integral does not converge, indicate that and move on. It is not appropriate to give a value for an integral that does not converge!

1. $\int_0^{\pi/2} \sin^4\theta \cos^6\theta \, d\theta$.

2. $\int_0^{\pi/2} \sin^5\theta \cos^6\theta \, d\theta$.

3. $\int_0^{\pi/2} \frac{\ln\sin\theta}{\sin\theta} d\theta$.

4. $\int_0^{\pi/2} \frac{\ln\sin\theta}{\cos\theta} d\theta$.

5. $\int_0^1 \frac{\ln^2(1-t)}{t\sqrt{t}} dt$.

6. $\int_0^1 \frac{\ln t \, \ln(1-t)}{t} dt$.

7. $\int_0^1 \frac{\ln(1-t)}{t^2\sqrt{t}} dt$.

8. $\int_0^1 \frac{t+\ln(1-t)}{t^2\sqrt{t}} dt$.